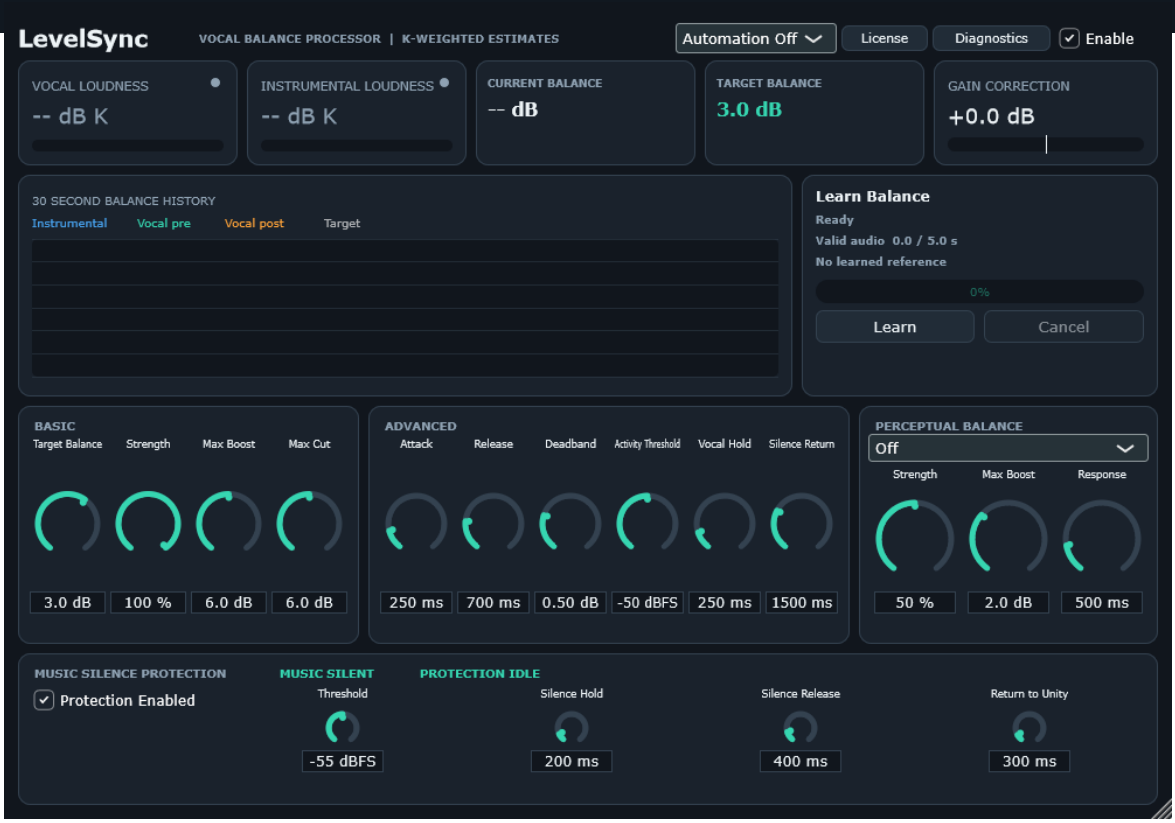
LevelSync
User Manual

Automatic vocal balancing for VST3 and Standalone



Version 0.5.2 | Windows x64 | November 2026

This manual documents the interface and behavior implemented in LevelSync 0.5.2.

1 Overview and quick start

LevelSync smoothly adjusts vocal gain against a stereo instrumental sidechain. The sidechain is used only as a reference and is never mixed into vocal output.

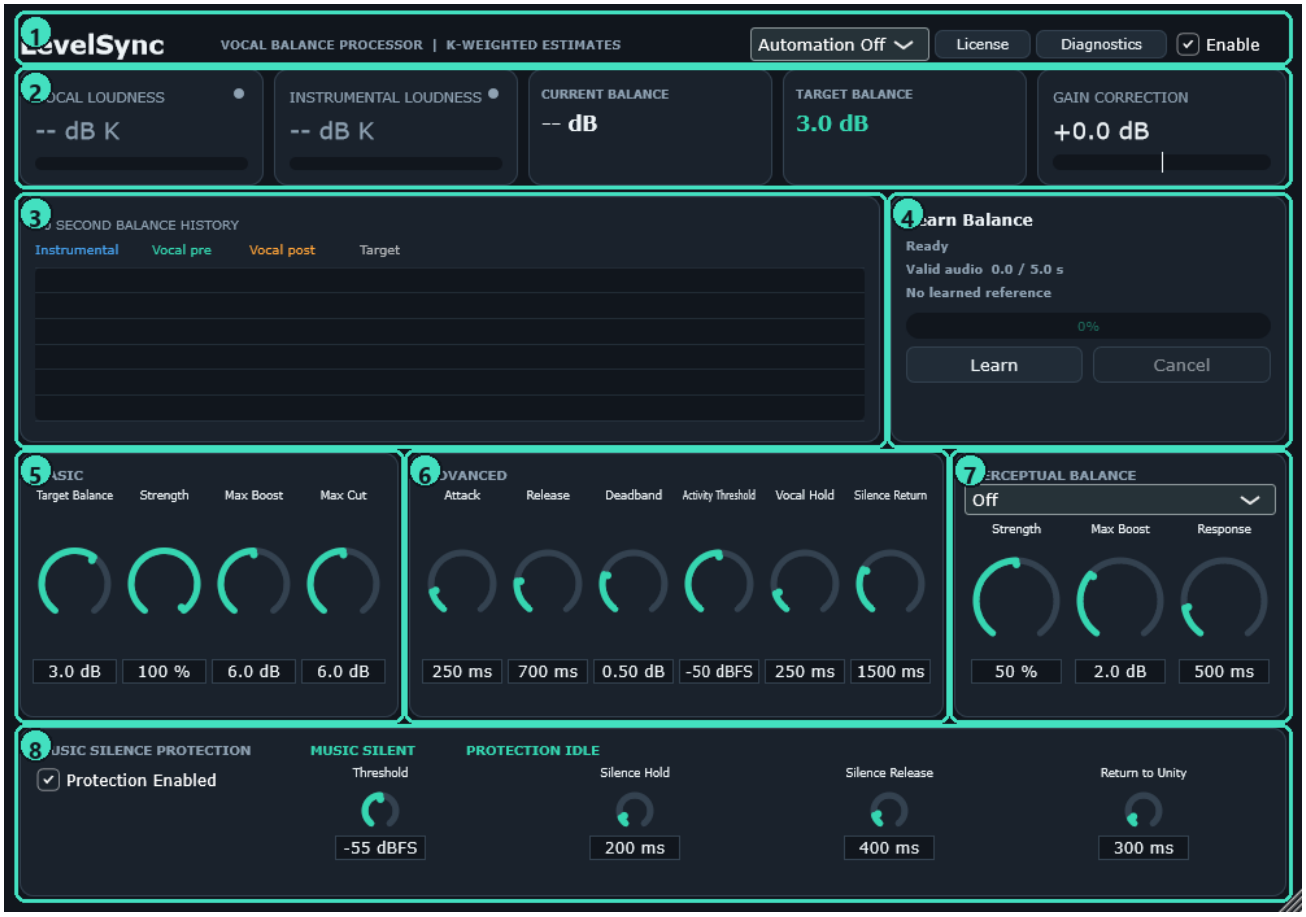
Signal flow

Vocal → loudness/activity → balance controller → gain smoothing → output. Instrumental sidechain → loudness, silence, and spectral analysis.

Quick start

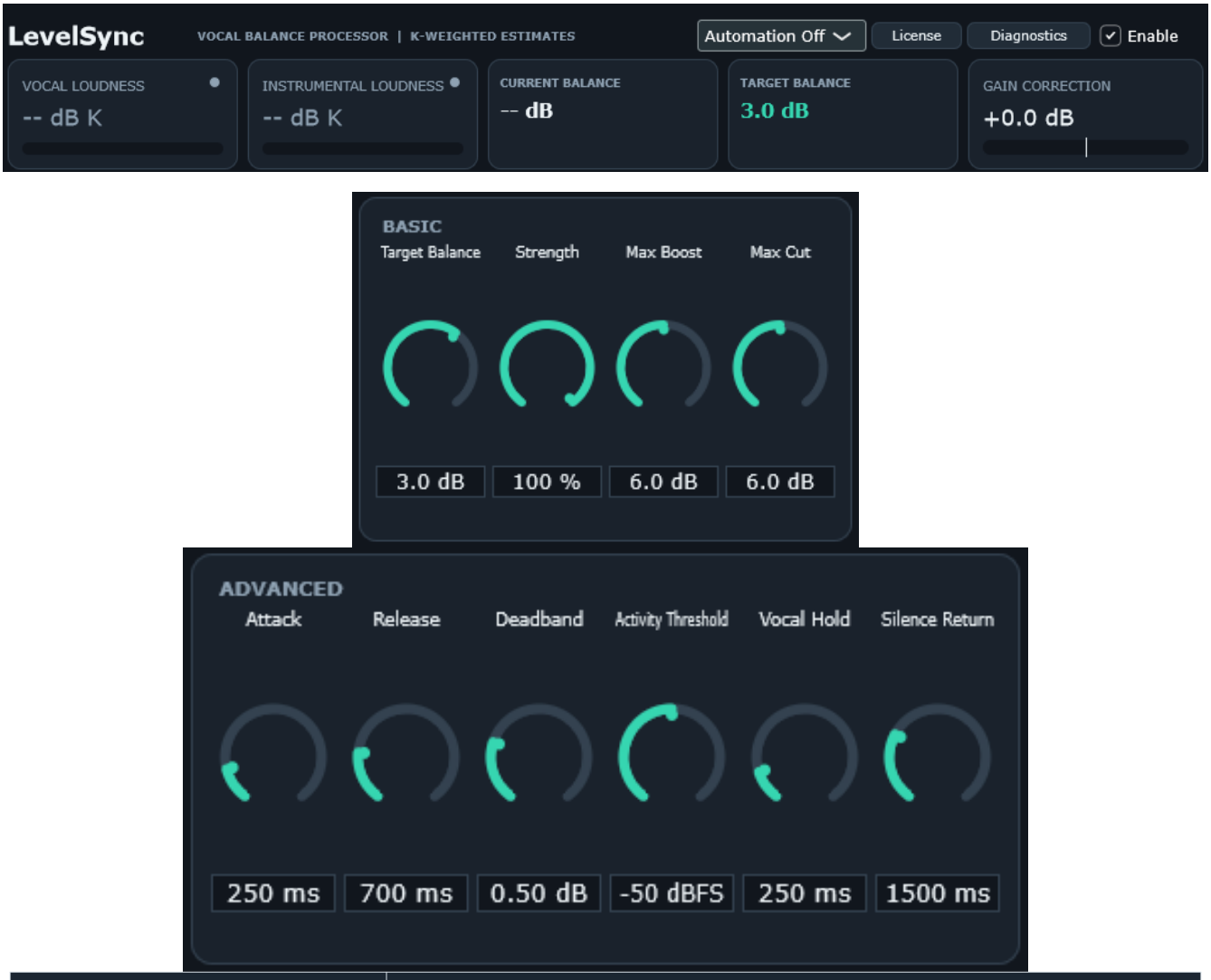
- Insert LevelSync on the vocal track.
- Route the instrumental track or bus to the stereo sidechain.
- Confirm valid vocal and instrumental meters.
- Set Target Balance and timing, or run Learn Balance.
- Choose Automation Write only when recording a Rider Gain curve.

2 Interface map



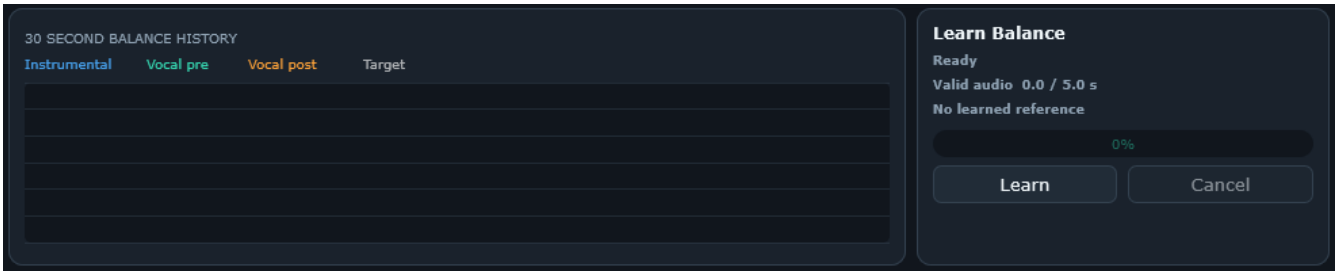
#	Area	Purpose
1	Header	Automation selector, License, Diagnostics, and Enable.
2	Monitoring	Vocal/music loudness, balance, and correction.
3	History	Thirty seconds of instrumental, vocal, and target.
4	Learn	Reference calibration.
5	Basic	Target, strength, and gain limits.
6	Advanced	Timing, deadband, activity, and return.
7	Perceptual	Masking-based boost.
8	Silence protection	Neutral return during music stops.

3 Monitoring and balance controls



Control	Function
Target Balance	Desired vocal position relative to music.
Strength	Scales automatic correction.
Max Boost / Cut	Limits positive and negative movement.
Attack / Release	Controls reduction and recovery speed.
Deadband	Prevents small unnecessary movements.
Activity Threshold / Hold	Detects useful vocal activity.
Silence Return	Returns to unity without reliable vocals.

4 Learn and perceptual processing



Learn Balance

Learn Balance returns gain to unity, then collects 5 seconds of valid vocal and music. Silence pauses progress without replacing the previous reference. Completion stores a robust balance and spectral reference.



Perceptual Balance

Perceptual Balance analyzes competition in 24 ERB bands. Gentle, Balanced, and Strong scale a positive masking correction that remains inside global Max Boost.

5 Music Silence Protection



Music Silence Protection uses separate unweighted dBFS energy. After Silence Hold, it suspends music-based correction and returns vocal gain to 0 dB. Music must remain 3 dB above Threshold for Silence Release before normal control resumes.

Control	Range / Default	Function
Threshold	-90..-20 dBFS / -55	Silence detection level.
Silence Hold	0..2000 ms / 200	Confirms silence.
Silence Release	0..3000 ms / 400	Confirms music return.
Return to Unity	50..3000 ms / 300	Smooth neutral return.

6 DAW automation: Write and Read

Mode	Behavior
Off	Normal internal processing. No automation is published.
Write	Applies automatic gain and publishes Rider Gain to the host at up to 25 Hz, with a 0.05 dB point threshold.
Read	Ignores automatic gain for output and follows the host Rider Gain curve with a 20 ms ramp.

Recording workflow

- 1. Expose the Rider Gain automation lane in the DAW.
- 2. Set the DAW lane to Write, Touch, or Latch.
- 3. Select Automation Write in LevelSync and play the section.
- 4. Stop, set the DAW lane to Read, and select Automation Read in LevelSync.
- 5. Edit, simplify, copy, or delete points with DAW tools.

Do not leave the lane in Write after recording. Host support varies; validate plugin-originated automation in your DAW. Standalone has no host and cannot record automation.

7 Diagnostics, installation, and troubleshooting

Diagnostics

Diagnostics displays pre-gain vocal/music estimates, current/target balance, error, target/applied gain, activity, sidechain validity, limits, masking confidence, and perceptual correction. Missing values mean invalid measurement, not simulated data.

Installation

- Copy the complete LevelSync.vst3 bundle to a folder scanned by the DAW.
- The standard Windows path is C:\Program Files\Common Files\VST3.
- Insert LevelSync on vocals and send music to the stereo sidechain.
- Rescan and confirm Instrumental Loudness before evaluating balance.

Troubleshooting

Situation	Suggested action
No sidechain	Check the stereo auxiliary route and DAW VST3 sidechain support.
Learn paused	Provide active vocals and music above the silence threshold.
Automation not recorded	Expose Rider Gain, arm host automation, and test Write/Touch/Latch.
Edited curve overwritten	Switch both the DAW lane and LevelSync to Read.
Pumping	Increase Release, Deadband, or Vocal Hold.

8 Complete parameter reference

Parameter	Unit	Minimum	Maximum	Default
Balance Offset	dB	-12	12	3
Strength	%	0	100	100
Max Boost	dB	0	12	6
Max Cut	dB	0	12	6
Attack	ms	50	1500	250
Release	ms	100	3000	700
Deadband	dB	0	2	0.5
Vocal Threshold	dBFS	-80	-20	-50
Vocal Hold	ms	0	2000	250
Silence Return	ms	100	5000	1500
Perceptual Mode	-	Off	Strong	Off
Perceptual Strength	%	0	100	50
Max Perceptual Boost	dB	0	6	2
Masking Response	ms	50	3000	500
Music Silence Threshold	dBFS	-90	-20	-55
Music Silence Hold	ms	0	2000	200
Music Silence Release	ms	0	3000	400
Music Return to Unity	ms	50	3000	300
Music Silence Protection	bool	Off	On	On
Enable	bool	Off	On	On
Automation Mode	-	Off	Read	Off
Rider Gain	dB	-12	12	0

9 Limitations, safety, and licensing

- No source separation; bleed and noise can affect detection.
- Positive gain requires downstream headroom.
- Automation is block-rate, not sample-accurate.
- Plugin-originated automation behavior varies by DAW.
- Automated tests do not replace real DAW validation.

Licensing

Original LevelSync code is AGPL-3.0-or-later. JUCE 9.0.5 is used under the AGPL option. See LICENSE, NOTICE, and THIRD_PARTY_NOTICES.md. No warranty is provided.

Downloads and source

https://github.com/ricferreiradev/levelsync/releases
https://github.com/ricferreiradev/levelsync

LevelSync v0.5.2